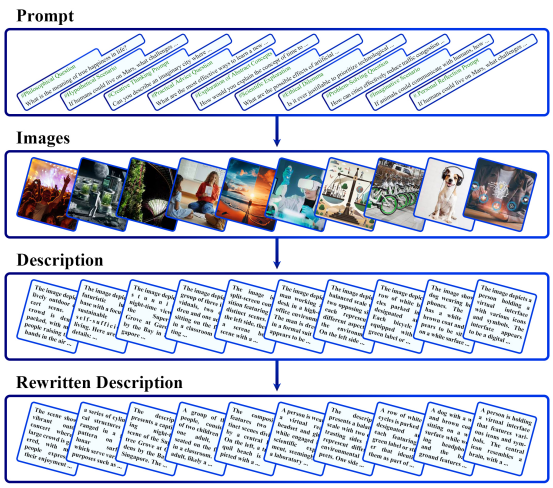

Figure 3: This figure illustrates the process of preparing image view embeddings and provides an example for 10 input prompts. Row 1 displays 10 prompts from various subjects. Row 2 shows images crawled based on each input prompt. Row 3 presents the descriptions generated for the images, and Row 4 contains the rewritten descriptions, which serve as our image view embeddings.

as novelty, diversity, or correctness—individually. In contrast, our work takes a comprehensive approach by considering all these aspects to assess the model's performance.

3.2.1 Diversity Measure

The diversity in the text generated by LLMs can be measured from various aspects. Two of the most important aspects are:

- Diversity across different text responses.
- Diversity within different tokens generated in a single text response.

In this work, first we try to describe each metric and then plot the importance of each one in diversity measurement and special cases for each metric and finally talk about the results.

MTLD: Lexical diversity is a key measure of linguistic richness in generated text. To evaluate this, we use the Measure of Textual Lexical Diversity (MTLD) (McCarthy and Jarvis, 2010). MTLD calculates the mean length of text segments that maintain a predefined type-token ratio (TTR). This approach overcomes limitations of traditional TTR metrics, which are sensitive to text length. Following (McCarthy and Jarvis, 2010), we set the TTR threshold to 0.72, as this value has been empirically validated to balance sensitivity and robustness across a variety of text datasets.

Semantic Diversity of Text (SDT) We measure semantic diversity using Term Frequency–Inverse Document Frequency (TF-IDF) representations of all responses. First, each response is transformed into a TF-IDF vector based on term usage and distinctiveness. Then, we compute the pairwise cosine similarity among these TF-IDF vectors to quantify how similar the responses are. Because higher similarity implies lower diversity, we define our overall semantic diversity metric as:

$$\text{SDT} = 1 - \overline{\text{Sim}},$$

where $\overline{\text{Sim}}$ is the average of all pairwise cosine similarities between the TF-IDF vectors of the responses. Consequently, larger values of SDT indicate greater variation in term usage—i.e., more semantic diversity. This formulation ensures that near-identical responses (high cosine similarity) yield low diversity scores, while dissimilar responses produce high diversity scores.

Semantic Diversity of Embeddings (SDE) BERT-based semantic diversity measures the variation in the *semantic content* of generated text. Unlike lexical diversity, which focuses on surface-level differences in word usage, BERT embeddings capture deeper contextual differences by mapping sentences into a dense semantic space. We compute pairwise cosine similarity among the embeddings of all responses and then take the average cosine distance (i.e., $1 -$ cosine similarity) to quantify diversity. Higher semantic diversity indicates a broader range of meanings and greater contextual richness. This approach leverages the contextualized representations of BERT (Devlin et al., 2019), offering a robust and nuanced assessment of the diversity and coherence of model-generated outputs. The **Semantic Diversity of Embeddings** (**SDE**) can be quantified by averaging the pairwise *cosine distance* between all embeddings. Given a set of embeddings $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$, the cosine similarity between two embeddings $\mathbf{e}_i$ and $\mathbf{e}_j$ is defined as: $\text{CosineSimilarity}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}$. The corresponding cosine distance is: $1 - \text{CosineSimilarity}(\mathbf{e}_i, \mathbf{e}_j)$. Thus, the overall **SDE** is:

$$\textbf{SDE} = \frac{2}{n(n-1)} \sum_{1 \le i < j \le n} \left(1 - \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}\right),$$

where n is the total number of embeddings (responses).

Self-BLUE Self-BLEU evaluates the diversity of generated text by measuring how similar each generated sample is to the others. Unlike traditional BLEU, which compares a generated sample to reference texts, Self-BLEU treats each sample as a reference for the others. A high Self-BLEU score signals redundancy, reflecting repeated patterns or limited variability (Zhu et al., 2018). In our setting we subtract the mean value across all answers from 1 to measure the diversity across all answers.

Lexical Entropy It quantifies the uncertainty or variability in word choice within generated text, serving as a measure of lexical diversity. Higher entropy indicates a broader and more varied vocabulary, while lower entropy suggests repetitive or predicTable word usage. We calculate Shannon entropy (in bits) over all tokens in the generated responses, based on their relative frequencies. For a given set of responses, the probability of each unique token is determined, and the entropy is computed as: $H = -\sum_i p_i \log_2 p_i$ where p_i represents the probability of the i-th token. A higher lexical entropy signifies reduced repetition and a richer vocabulary, making it a valuable indicator of linguistic creativity in language models.

3.2.2 Novelty Measure

Novelty detection is critical for assessing large language model (LLM) outputs, as it differentiates existing knowledge from newly introduced information. This task can be framed as Natural Language Inference (NLI), wherein one determines whether a hypothesis contains content not found in a premise set. Prior studies have employed both NLI-based methods and embedding techniques such as SBERT (Ghosal et al., 2022) to measure semantic similarity, illustrating the importance of novelty detection for refining LLM-generated text.

In our work, we wanted to assess the inherent capability of LLMs to perform as NLI models in detecting novel versus non-novel documents. To achieve this, we used **TAP-DNLD 1.0** (Ghosal et al., 2018) a dataset consisting of around $2.8k$ novel and $2.7k$ non-novel documents, and each document belongs to one of the ten categories and each target document labeld versus three source document by human. For comparison , first we sampled from this dataset in three different seeds and in each seed from each category sampled 5 novel and 5 non-novel document to have a balance data and totally 300 document then We used GPT-4o and SBERT as novelty detector to have better comparison and results reported in Table 1.

Seed	Model	Accuracy	Precision	Recall	F1-score
Seed0	SBERT	0.6700	0.6269	0.8400	0.7179
	GPT-4O	**0.6900**	**0.6415**	**0.8600**	**0.7347**
Seed1	SBERT	**0.7200**	**0.6528**	**0.9400**	**0.7705**
	GPT-4O	0.6600	0.6086	0.9000	0.7260
Seed2	SBERT	0.6900	0.6338	0.9000	0.7438
	GPT-4O	**0.6950**	**0.6368**	**0.9100**	**0.7492**

Table 1: Classification metrics to evaluate novelty detectors GPT-4o and SBERT models on dataset **TAP-DNLD 1.0** (Ghosal et al., 2018).

After establishing the baseline performance of these models, we utilized GPT-4o and SBERT as novelty detectors to evaluate the novelty of generated outputs across different language models, as well as to assess the impact of our method on the novelty of generated responses. The novelty detection process is illustrated in Figure 4. In this approach, for each set of prompt outputs, the first generated response is considered novel by default. Subsequent responses are sequentially analyzed by the novelty detector, where each new answer is compared against previously identified novel responses. This comparison is conducted in a hypothesis-premise framework, where the new response serves as the hypothesis, and previously identified novel answers form the premise set. If the new answer introduces additional information not present in the premise set, it is classified as a novel answer and added to the premise set. Otherwise, it is labeled as redundant.

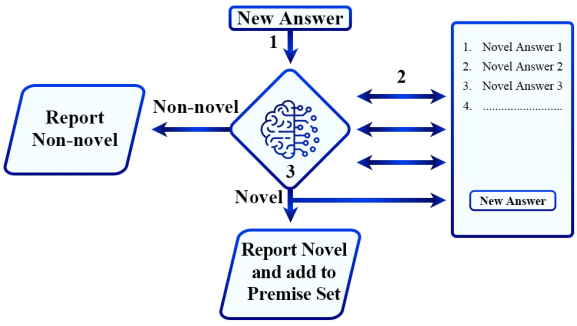

Figure 4: Novelty detection method by using GPT-4o and SBERT.

3.2.3 Correctness Measure

Answers could be novel or even diverse but completely irrelevant to the input prompt (see example in appendix A), or they can have different structures. In this section we want to answer these two questions:

- Is the generated answer correct and relevant to the given input prompt?
- How well does the generated answer adhere to proper language structure and grammatical accuracy?

To address the first question, we designed an experiment to assess whether language models can accurately detect relevant answers to a given prompt. For this evaluation, we used the GPT-WritingPrompts dataset (Huang et al., 2024), which contains approximately $97k$ unique prompts along with responses from both humans and GPT-3.5. However, in our study, we exclusively utilized the human-generated answers. We sampled data using ten different random seeds, selecting 1K samples per seed. These samples were then clustered into ten distinct groups. From each cluster, we randomly picked 100 prompts. For each selected prompt, we assigned one relevant answer from the same cluster, labeling it as "Correct," and one irrelevant answer from a different cluster, labeling it as "Incorrect," ensuring that the chosen answers were distinct. Since the answers are approximately 500 words long, we assume that if an answer is related to the prompt, its summary will also be relevant, and similarly, if an answer is irrelevant, its summary will remain unrelated. Based on this assumption, we use the GPT-4o model to generate summaries for both correct and incorrect answers, setting the maximum length to 250 words. Next, we evaluate the ability of two well-known LLMs, GPT-4o and DeepSeekV3 (DeepSeek-AI et al., 2024), to determine the correctness of the answers. The results, presented in Table 2, indicate that GPT-4o outperforms DeepSeekV3 in detecting correctness. An example prompt and its related responses are provided in Appendix B. Based on these findings, we use GPT-4o for correctness detection in our experiments.

Model	Accuracy	Precision	Recall	F1-score
DeepSeekV3	0.8410 ± 0.0223	**0.9564 ± 0.0285**	0.7270 ± 0.0447	0.8191 ± 0.0301
GPT-4O	**0.8980 ± 0.0214**	0.9468 ± 0.0255	**0.8310 ± 0.0316**	**0.8946 ± 0.0228**

Table 2: Results on classification task to detect correct answer across all 10 seeds for DeepSeekV3 and GPT-4o models. Complete results can be found in Table 13, Appendix A.

To address the second question, we designed an experiment to evaluate language models based on their ability to assign scores to generated English texts. For this evaluation, we utilized the IELTS Writing Task 2 dataset with labeled scores (Mazlumi, 2023) from Kaggle. This dataset contains 642 responses for Task 1 and 793 responses for Task 2. Since only Task 2 is relevant to our study, we used its questions and answers to assess language models. We randomly sampled 100 responses in three different seeds, conducting two iterations to evaluate text quality. Two well-known models assigned scores between 1 and 9 (based on the IELTS writing score range), and the results are presented in Table 3. Based on this experiment, we selected the DeepSeekV3 model to evaluate responses in terms of grammar and overall English structure, assigning each answer a score between 1 and 10. The prompt templates for both experiments are provided in Appendix B. Based on this experiment we used DeepSeekV3 model to evaluate the outputs.

Seed	GPT-4o	DeepSeekV3
Seed0	2.3775 ± 0.0350	**2.0775 ± 0.0025**
Seed1	2.2288 ± 0.0512	**2.0300 ± 0.0050**
Seed2	2.1562 ± 0.0587	**1.9050 ± 0.0500**

Table 3: MSE (Mean ± Std) for GPT-4o and DeepSeekV3 per seed for two iterations. Lower values are bolded.

4 Experimental Results

In Section 3.2.1, we introduced various diversity metrics. In this section, we provide experimental results to assess the diversity of different language models. Additionally, we applied the multi-view embedding method to four open-source models—GPT-2 Medium (Radford et al., 2019), Qwen2.5-1.5B (Team, 2024), Llama 3.2-3B (MetaAI, 2025) ,and DeepSeek-R1-7B (DeepSeek-AI et al., 2025)—as well as two API-based models, GPT-4o (OpenAI, 2023) and GPT-4o Mini (OpenAI, 2024)[1]. Our experiments utilized 10 diverse prompts spanning multiple domains (provided in Appendix B), instructing each model to generate responses. This process was repeated across different sample sizes ranging from 100 to 2000, with a fixed maximum sequence length of 125, resulting in a total of 909500 generated responses and for having fair compression we used the same parameters for all models like $tempereture = 0.9$ and $top_k = 0.95$. The generated outputs were then evaluated in terms of diversity, novelty, and correct-

[1]We conducted all expermiments on one A100 GPU and spent 750$ for API models.